

Countable Information, Uncountably Many Boxes

Sharp measurable prediction bounds, exact decision-tree enumeration, and an all-orders asymptotic expansion

Prepared for Vladimir Reshetnikov

October 3, 2026

Abstract

We study the distinction between choosing an unread coordinate and assigning a measurable event to a correct prediction. For a nonempty standard Borel set of boxes, a finite alphabet of size q , and m deterministic players whose output maps are measurable for the cylinder σ -algebra, we prove that the optimal guaranteed number of correct guesses is exactly $\lfloor m/q \rfloor$. The proof does not assume that success events are measurable. A simultaneous measure-extension construction supplies fair success probabilities for any countable collection of such players. Conversely, measurability of a blind player's success event forces its target range to be countable, pointwise or outside a cylinder-null set according to the chosen completion.

The finite problem leads to a separate structural result. Blind binary output maps correspond to labeled perfect matchings of the hypercube, whereas executable maps correspond precisely to recursively sliceable matchings. We derive an exact inclusion–exclusion recurrence for the latter, prove that the two classes first differ at four boxes, and establish

$$F_n = \frac{e^{\gamma 2^n}}{n} \left(1 - \frac{3}{2n} + \frac{17}{4n^2} - \frac{12}{n^3} + O(n^{-4}) \right),$$

with a rational algorithm for every further asymptotic coefficient. We also prove an explicit doubly exponential upper bound on the fraction of blind maps that are executable. Finite checks, an oracle counterexample-extraction theorem, and a Busy Beaver obstruction connect the analysis to the certificate-oriented architecture of ProveIt.

Scope and status. All stated theorems have proofs in this note. The finite computations were executed independently in Python. The infinite results have not been checked in Lean or Rocq, and historical priority is not established. The restricted measurable problem is settled here; Glazer's unrestricted question about what ZF proves is not settled here. Neither authorship nor endorsement by Elliot Glazer is implied.

Contents

1	Research question and contribution map	2
2	Configurations, blindness, and measurable output	5
3	The sharp measurable-output bound	6
4	A simultaneous fair extension of probability	7
5	Measurable success forces a countable target set	8
6	A continuous selector with maximally nonmeasurable success	9
7	Finite output maps and executable decision trees	10
8	Exact enumeration of executable maps	12
9	All-orders asymptotics for sliceable matchings	13
10	Executable blind maps are doubly exponentially rare	16
11	How much does a coordinate budget reveal?	17
12	Effective finite certificates and the Busy Beaver barrier	18
13	Executed checks and reproducibility	19
14	Foundations and a concrete ProveIt integration plan	20
15	Further research questions and topics	21
16	Conclusion	23
A	Assumption and claim ledger	23
B	Source and priority audit	24

1 Research question and contribution map

Glazer studies prediction games in which players inspect boxes and then guess an unopened value. In Section 5 of [1], he asks whether ZF proves that three players with continuum many binary boxes can guarantee at least two correct guesses. His work on weak choice principles [2] and the finite and infinite puzzles described in his research profile [3] make regularity, executability, and explicit axiomatic boundaries natural questions to investigate.

The relevant ProveIt materials are its Busy Beaver machine interface, finite-list coding in arithmetic, and explicit first-order set-theoretic developments [6, 7, 8, 9]. Rather than importing an unverified mathematical claim from those projects, we use their architecture: distinguish a semantic statement from an executable certificate, and distinguish both from the theorem that validates the certificate. The repository snapshot used throughout is 6fef5383b126be343cccc9baf47081ef37ab5afe.

The central issue is not a numerical estimate. On an uncountable product, the following statements are different:

- (i) the rule selecting a box and a guess is measurable;
- (ii) the event that the selected box contains the guess is measurable;
- (iii) the rule can actually be implemented without inspecting its final target.

None of these distinctions can safely be suppressed. We obtain two complementary lines of results.

1.1 The continuum line

Theorem 3.1 gives the exact measurable-output minimax value, including the three-player binary case. Theorem 4.1 constructs a simultaneous fair extension of the product probability for countably many players. Theorem 5.1 shows that measurable success is a much stronger restriction on the range of targets than measurable output. The diagonal example in Section 6 makes the difference explicit: its output rule is continuous, but its success event has inner measure zero and outer measure one. We classify every extension obtained by adjoining that event.

1.2 The finite line

Theorem 7.3 characterizes executable binary output maps by a recursive property of hypercube matchings. The sequence F_n counting the underlying executable matchings begins

$$1, 2, 9, 232, 206065, 212181312096, \dots$$

It is *not* the sequence of all perfect matchings, OEIS A005271 [5]; that sequence already differs at $n = 4$. We prove an exact recurrence, a full asymptotic expansion, and an explicit rarity bound. These results are not asserted to be absent from the literature or the OEIS: exact-sequence searches did not yield a usable classification during this investigation.

1.3 What is, and is not, claimed as a contribution

The countable-support lemma, finite-coordinate averaging, Radon–Nikodym theorem, compactness of a finite-alphabet sequence space, and halting-time diagonalization are standard ingredients. The contribution developed here is their precise application to the output/success distinction, together with the decision-tree

classification, enumeration, asymptotic calculation, and executable finite checks. The measure-extension classification is a special-case application of the classical theory represented by Łoś and Marczewski [4], not a claim to have invented measure extension.

These statements constitute a research note with complete proofs and testable finite consequences. They are not presented as a verified historical breakthrough or as a resolution of an unrestricted choiceless-set-theory problem.

1.4 Provenance and place in the collection

[write] Added 3 October 2026, batch 87. This report is a single manuscript, printed in full: manuscript 02 of batch 87 of ProveIt’s incoming-reports intake, delivered as the archive `glazer_box_games_research.zip` (inner directory `glazer_box_games/`, main file `article.tex`, a 24-page US Letter PDF) in the arrival commit `fa0a0576e`, and placed in commit `d151b39ca` in the research-report collection as `ordinals-and-order-types/measurable-box-games`. The author line reads “Prepared for Vladimir Reshetnikov” and carries no AI wording; like the other deliveries of the intake, the report is AI-assisted, unrefereed and not formalized. Its five sister manuscripts of batch 87 concern Glazer’s *other* question paper, *A Topological Tennenbaum Theorem* (arXiv:2311.13699), and became Parts VI–IX of the surreal-collection report `Algebra/SurrealNumbers/docs/foundations-and-computation/polis-h-models-of-omnific-arithmetic`; this manuscript shares no theorem, question or notation with them. Being a single source, the report needed no merge choices. The write prefixed all 61 delivered labels with `mbg:` and updated every reference; added the four labels `mbg:sec:provenance`, `mbg:sec:notation`, `mbg:app:ledger` and `mbg:app:audit`; added these dated **[write]** notes, the title-page line, the `pageanchor` switch around the title page (a delivered build gave a duplicate `page.1` destination) and a dated addition to the bibliography entry [1]. No statement, proof, number or non-claim of the manuscript was changed.

Pin and repository claims. The pin `6fef5383b126be343cccc9baf47081ef37ab5afe` (`6fef5383b`, quoted throughout) is the ProveIt commit the manuscript was written against; it is an ancestor of the placement. The three repository files it read, `Computability/BusyBeaver/Lean/BusyBeaver/Core.lean`, `Logic/PeanoArithmetic/ListCoding/README.md` and `SetTheory/ClosureAxiomatization/README.md`, were byte-identical at the pin and at the write (blobs `4961655e`, `74164ac6`, `fb0ab511`), and the descriptions of them in Sections 12 and 14 are accurate: the Busy Beaver core’s opening comment separates the blank-tape machines and the maximum property of the score function from a separately stated compiler fact; the list-coding README documents validity guards, unique decoding and round trips; the closure-axiomatization README separates its semantic directions from the deductive-equivalence layer. At placement the intake searched the whole tracked tree for box, hat, guessing and prediction games, for `2211.10474` and for `A005271`, and found nothing: no report or formal project in ProveIt treats this question, and the report continues no repository material. Its place in the collection confers no formal status; no statement of it is formalized in Lean or Rocq.

Glazer’s paper. The arXiv record of [1], read on 3 October 2026, lists a single version, v1 of 15 November 2022 (6 pages; math.LO, math.CO; DOI `10.48550/arXiv.2211.10474`); the date line “January 9, 2023” that the manuscript’s “inspected PDF” carries is printed in arXiv’s PDF of that same v1, not a revision. The question of Section 5 (“More box games”) of that paper reads “Does ZF prove that $G(\{3\}, \mathbb{R}, \{0, 1\})$ is winnable?”, where winning means that at most one of the three players guesses wrong; it is Research question Q1 of Section 15. **It is not the “Glazer’s Question 1” of the surreal report above**, which is Question 1 of the Topological Tennenbaum paper, about the reverse-mathematical strength of Glazer’s Theorem 2 there.

Review in the Hilbert’s-tenth research tree. Before placement, another session reviewed all six batch-87 archives at their arrival revision (commit `f15962acd`): `Computability/HilbertTenthProblem/Papers/research-h-wip/native-stream-queue/review_baire_polish_arithmetic_intake.md`, with an independent cross-reading in `review_baire_polish_arithmetic_intake_independent.md` beside it and a receipt in `review_baire_polish_arithmetic_intake.json`. For this manuscript it read the delivered lines 64–73, 82–110, 386–428, 776–820 and 927–955 (the abstract; Section 1 through Section 1.3; Section 7 from its start through Lemma 7.4; Section 12; the conclusion and the ledger of Appendix A). It confirms the statements as scoped here: legal finite binary decision trees correspond to recursively sliceable matchings and blind output tables need not be executable; the extraction of Theorem 12.1 is a partial procedure for which the totality and legality promise is essential; no computable

horizon bound from program length exists, and the Busy Beaver comparison is conditional on an encoding and an additive-overhead compiler. It records that these results give no fixed-size Diophantine history certificate and no paid arithmetic schedule, so they leave the Hilbert’s-tenth programme’s 84-operation universal polynomial unchanged, and that the probability, matching-enumeration and asymptotic claims were *not* reviewed there. Nothing in the review required a change of scope here.

OEIS, checked at intake. On 3 October 2026 the intake searched the OEIS for 1, 2, 9, 232, 206065, for 9, 232, 206065 and for the ternary 2, 21, 33976: none matches an entry. The entry A005271, “Number of perfect matchings in n -cube”, begins 1, 2, 9, 272, 589185, 16332454526976, $\dots$, agreeing with $M_1, \dots, M_5$ of Section 8. This is a check of the database on that date, not of the literature, so the manuscript’s own reservation in Section 1.2 stands. Nothing was submitted to the OEIS, by the manuscript’s author or by the intake; any submission is a human editor’s decision after review. An independent program at intake reproduced $F_1, \dots, F_6$, the ternary $F_1^{(3)}, \dots, F_5^{(3)}$, D_6 , the integer comparison of Theorem 10.1, the 232 sliceable and 40 nonsliceable matchings of Q_4 by brute force, and the two decimal enclosures of γ .

Neighbouring reports. None shares a theorem. The games of `ordinals-and-order-types/games-on-ordinals/` are different objects: in `open-query-membership-games` a Seeker asks open-set queries about a hidden point on a topological space (its transcripts and finite-budget certificates are adaptive query strategies, but nobody guesses an unread coordinate); `point-separating-game-values` and `ordinal-chomp-transition-at-two` concern point-separating games and ordinal Chomp. `ordinals-and-order-types/non-baire-translation-invariant-ideal` is the nearest in spirit: it builds, in ZFC from a free ultrafilter, an ideal without the Baire property, the kind of choice-built irregular object whose absence Research question Q2 asks about for the Baire-property version of Theorem 3.1.

1.5 Notation used in this report

[write] **Added 3 October 2026, batch 87.** The manuscript reuses several letters with local meanings. None was renamed; this table fixes the readings, with the tempting false one where confusion is likely.

Symbol	Meaning (and what it is not)
γ	the growth constant $\lim 2^{-n} \log F_n \approx 0.438158$ of (11); <i>not</i> Euler’s constant $0.5772\dots$
F_n, M_n	sliceable (executable) matchings of Q_n ; all perfect matchings (A005271). They first differ at $n = 4$ ($232 < 272$).
B_n, L_n	blind and legal binary output maps, $2^{2^{n-1}} M_n$ and $2^{2^{n-1}} F_n$; B_n has nothing to do with a Busy Beaver function
$\text{BB}_{\text{time}}, H$	a program-length running-time Busy Beaver function and the maximal horizon (Section 12); <i>not</i> Rado’s n -state score function formalized in Computability/BusyBeaver. In Section 11, H is entropy.
I	the set of boxes; in Section 11 also mutual information $I(Z; X_K)$
$\mathbb{E}, E_\tau, E$	expectation; the success set of τ ; the diagonal success set of Section 6
C	a group of players sharing an external target (proof of Theorem 3.1); Cantor space $2^{\mathbb{N}}$ (Section 6); the formal series $C(z)$ of (23)
D	the integer score $D(u)$ (proof of Theorem 3.1); the count D_6 of (25); an event in the proof of Theorem 11.1
r	list size (Corollary 3.3); the ratio $r_n = F_n/F_{n-1}^2$; the label $r = x \upharpoonright \mathbb{N}$ in Section 6
$R, \mathbb{R}$	the formal series $R(z)$ of (21); the real line
T, L	the replacement map $T(Y, U)$ (Theorem 4.1); the time and prefix-length bounds T, L in the proof of Theorem 12.1 (L is not L_n)
Q, Q_n, q	the alphabet $\{0, \dots, q-1\}$; the cube graph; the alphabet size (not a q -series variable)
Σ_I	the cylinder σ -algebra, <i>not</i> the Borel σ -algebra of the product topology and not its completion
Q1–Q12	Research questions of Section 15; Q1 is Glazer’s box-game question, not the “Question 1” of his Topological Tennenbaum paper

2 Configurations, blindness, and measurable output

Fix $q \geq 2$, put $Q = \{0, \dots, q-1\}$, and let I be a nonempty set. A *configuration* is a function $x : I \rightarrow Q$. Write

$$X_I = Q^I, \quad \Sigma_I = \sigma\{x : x_i = a \mid i \in I, a \in Q\}.$$

The probability measure μ_I on Σ_I is the product measure under which every fixed finite family of coordinates is independent and uniform on Q .

Remark 2.1 (Which σ -algebra?). The cylinder σ -algebra Σ_I is not, by definition, the full Borel σ -algebra of the uncountable product topology. Nor is it automatically its own completion. Every measurability assertion below names the relevant σ -algebra. This is essential, not a technical convention that can be changed without affecting the theorems.

For $i \in I$ and $v \in Q$, let $x^{i \leftarrow v}$ denote the configuration obtained by replacing x_i by v and leaving all other coordinates unchanged.

Definition 2.2 (Blind output map). An output map is a total function

$$\tau = (b, a) : X_I \longrightarrow I \times Q.$$

It is *blind* if, whenever $\tau(x) = (i, c)$,

$$\tau(x^{i \leftarrow v}) = (i, c) \quad \text{for every } v \in Q. \quad (1)$$

Its success set is

$$E_\tau = \{x : x_{b(x)} = a(x)\}.$$

For a team $\tau_1, \dots, \tau_m$, the score is $S(x) = \sum_{p=1}^m \mathbf{1}_{E_{\tau_p}}(x)$.

Every total deterministic legal probing strategy induces a blind output map. Indeed, altering only the value of its ultimately unopened box changes none of the observations along the original run. Induction on the successive moves gives the same history and the same final output. This remains a necessary condition when a move reveals an entire set of boxes or histories are transfinite. We do not identify blindness with legal executability; Section 7 gives a finite counterexample to that identification.

From now on, for the continuum theorems, I carries a standard Borel structure. Measurability of τ means measurability from (X_I, Σ_I) to $I \times Q$ with its product Borel structure. In particular, singleton targets, equality of two target maps, and membership in any countable set of targets are measurable.

Definition 2.3 (Coordinate support). A set $A \subseteq X_I$ is *supported by* $J \subseteq I$ if membership in A is unchanged when coordinates outside J change. A function is supported by J if two configurations agreeing on J have the same function value.

Lemma 2.4 (Countable-coordinate dependence). *Every member of Σ_I has a countable support. Every Σ_I -measurable map to a countably separated measurable space has a countable support. Consequently a finite or countable family of measurable maps to standard Borel spaces has a common countable support.*

Proof. The sets possessing a countable support form a σ -algebra: complements keep the same support, and a countable union is supported by the union of its chosen supports. This σ -algebra contains each coordinate cylinder, proving the first assertion.

For a measurable function f , choose a countable family (C_k) of measurable sets separating points in its codomain. Choose a countable support J_k for $f^{-1}(C_k)$ and put $J = \bigcup_k J_k$. If $x \upharpoonright J = y \upharpoonright J$, then $f(x)$ and $f(y)$ belong to exactly the same C_k , hence are equal. A further countable union gives the last assertion.

The factorization is literal, not merely almost everywhere. Fix the symbol 0 outside J and let $s_J : Q^J \rightarrow Q^I$ be the corresponding section. Then s_J is measurable, $f_J = f \circ s_J$ is measurable, and $f = f_J \circ \pi_J$. $\square$

The lemma is proved here in the ordinary classical setting. The selection of countably many supports is among the steps that need a separate audit before attributing this proof to a weak choiceless base theory.

3 The sharp measurable-output bound

Theorem 3.1 (Exact minimax value). *Let I be a nonempty standard Borel space and let $m \geq 1$. For every team of m blind cylinder-measurable output maps there is a configuration x such that*

$$S(x) \leq \lfloor m/q \rfloor. \quad (2)$$

There is a legal team of constant output maps guaranteeing at least $\lfloor m/q \rfloor$ correct guesses on every configuration. Thus the optimum worst-case score in the measurable-output class is exactly $\lfloor m/q \rfloor$.

Proof. Choose a common countable support J for the outputs. On Q^J write the factored outputs as $(b_p(u), a_p(u))$. We separate targets inside J from targets outside J .

For $i \in J$ and $c \in Q$, the measurable fiber

$$A_{p,i,c} = \{u : b_p(u) = i, a_p(u) = c\}$$

is invariant under changing u_i , by blindness. A product integration in that one coordinate gives

$$\mu_J(A_{p,i,c} \cap \{u : u_i = c\}) = \frac{1}{q} \mu_J(A_{p,i,c}). \quad (3)$$

Therefore the internal score

$$S_{\text{in}}(u) = \sum_{\substack{1 \leq p \leq m \\ b_p(u) \in J}} \mathbf{1}_{\{u_{b_p(u)} = a_p(u)\}}$$

is measurable and satisfies

$$\mathbb{E} S_{\text{in}} = \frac{1}{q} \sum_{p=1}^m \mu_J(b_p \in J). \quad (4)$$

For each fixed u , group the remaining players according to their common target outside J . If C is one such group, let $n_{C,v}(u) = |\{p \in C : a_p(u) = v\}|$. Give the shared target the least symbol minimizing $n_{C,v}(u)$. Its contribution to the score is then

$$\min_{v \in Q} n_{C,v}(u) \leq |C|/q.$$

Define the integer-valued function

$$D(u) = S_{\text{in}}(u) + \sum_C \min_{v \in Q} n_{C,v}(u).$$

There are only finitely many players. The grouping is determined by finitely many measurable equality tests among their targets, so D is measurable. From (4),

$$\mathbb{E}D \leq \frac{1}{q} \sum_p \mu_J(b_p \in J) + \frac{1}{q} \sum_p \mu_J(b_p \notin J) = \frac{m}{q}.$$

An integer-valued function everywhere larger than $\lfloor m/q \rfloor$ would have expectation strictly larger than m/q . Choose u with $D(u) \leq \lfloor m/q \rfloor$. Complete u using the selected minimizing symbols at the finitely many external targets and the default symbol elsewhere. Since all outputs depend only on J , their values are unchanged. The resulting score is exactly $D(u)$.

For the reverse bound, fix a single box and divide the players' constant guesses as evenly as possible among the q symbols. Every symbol receives at least $\lfloor m/q \rfloor$ guesses. The players need not open any box, so this is a legal strategy. $\square$

Corollary 3.2 (The regular three-player continuum problem). *Three binary players with continuum many boxes and blind cylinder-measurable output maps cannot guarantee two correct guesses. They can guarantee one.*

This is a negative answer for a specified regularity class. It is not a statement that ZF disproves the existence of unrestricted winning strategies.

Corollary 3.3 (Lists of guesses). *Suppose every player supplies a set of exactly r possible symbols, where $1 \leq r < q$, and success means that the target value belongs to the list. With the corresponding blindness condition on the entire output (b, A) , the exact optimum guaranteed score is*

$$\lfloor mr/q \rfloor.$$

Proof. The internal fiber average in (3) is r/q . For an external target with t players, summing the number of containing lists over all symbols gives tr , so some symbol is in at most tr/q lists. The previous proof applies.

For sharpness, have all players choose the same box and let player p , indexed from zero, use

$$A_p = \{pr, pr + 1, \dots, pr + r - 1\} \pmod{q}.$$

Each list has r distinct symbols, and the mr consecutive residues distribute as evenly as possible among Q . $\square$

4 A simultaneous fair extension of probability

The preceding proof carefully avoided integrating the possibly nonmeasurable score on X_I . There is also a constructive way to enlarge the measurable space so that every player in a countable team has the expected fair marginal.

Theorem 4.1 (Independent replacement at external targets). *Let $(\tau_p)_{p \in \mathbb{N}}$ be a countable family of blind cylinder-measurable output maps on Q^I , with I standard Borel. There is a countably additive probability measure ν on*

$$\sigma(\Sigma_I, E_{\tau_0}, E_{\tau_1}, \dots)$$

such that $\nu \upharpoonright \Sigma_I = \mu_I$ and

$$\nu(E_{\tau_p}) = 1/q \quad (p \in \mathbb{N}).$$

For a finite team the same conclusion holds with just its success events adjoined.

Proof. Choose a common countable support J . Work on the auxiliary product space carrying an independent pair

$$Y \sim \mu_I, \quad U = (U_0, U_1, \dots) \sim \mu_{\mathbb{N}}.$$

All output pairs $(b_p(Y), a_p(Y))$ depend only on $Y \upharpoonright J$.

For each external target that occurs, use the first player selecting it as its representative. More precisely, when $b_p(Y) \notin J$, let

$$\rho_p(Y) = \min\{r \leq p : b_r(Y) = b_p(Y)\}.$$

Define a new configuration $T(Y, U)$ by leaving the coordinates of J unchanged, assigning the value $U_{\rho_p(Y)}$ to each selected external target $b_p(Y)$, and leaving all other coordinates equal to those of Y . Repeated targets receive the same replacement value, so the definition is consistent.

For each fixed coordinate i , T_i is measurable. Indeed, whether i is a selected target and which player is its first representative are countable combinations of measurable target-equality tests. The default case is also measurable. Hence T is measurable into (X_I, Σ_I) .

We next check the fixed-coordinate law. Conditional on $Y \upharpoonright J$, any fixed finite collection of coordinates outside J is transformed by replacing some of its entries with distinct independent uniform U coordinates. The selection is determined by $Y \upharpoonright J$, not by those outside values. Untouched entries remain independent uniform values of Y , independent also of U . Thus every fixed finite coordinate law of T is the original product law. Uniqueness of the probability measure on the generated cylinder σ -algebra gives

$$\mathbb{P}(T \in A) = \mu_I(A) \quad (A \in \Sigma_I).$$

This argument can equivalently be written by partitioning Q^J into the countably many possible representative patterns for the specified finite set, avoiding any disintegration on an uncountable product.

All outputs are unchanged by T , since it fixes J . If player p selects an external target, its pulled-back success event is

$$\{b_p(Y) \notin J, U_{\rho_p(Y)} = a_p(Y)\},$$

which is measurable and has conditional probability $1/q$ on that case. If it selects an internal target, its success probability is the sum in (3). Altogether,

$$\mathbb{P}(T \in E_{\tau_p}) = \frac{1}{q} \mathbb{P}(b_p \notin J) + \frac{1}{q} \mathbb{P}(b_p \in J) = \frac{1}{q}.$$

Finally, the collection of all subsets $A \subseteq X_I$ for which $T^{-1}(A)$ is measurable is a σ -algebra containing both Σ_I and every success event. The formula $\nu(A) = \mathbb{P}(T \in A)$ defines a countably additive probability on this σ -algebra, and restriction gives the required extension. $\square$

Remark 4.2. The theorem establishes existence, not uniqueness. It does not assert that success events already belong to the original product σ -algebra, or that every extension must give probability $1/q$. Section 6 exhibits the strongest possible nonuniqueness for one player. It also does not assert that the successes of different players are independent.

5 Measurable success forces a countable target set

Theorem 5.1 (Pointwise and completed target collapse). *Let I be any set and let $\tau = (b, a)$ be a blind output map on Q^I . No measurability of τ is assumed.*

(a) *If $E_{\tau} \in \Sigma_I$, there is a countable $J \subseteq I$ such that $b(x) \in J$ for every x .*

(b) If E_τ belongs to the μ_I -completion of Σ_I , there are a countable $J \subseteq I$ and a set $N \in \Sigma_I$ with $\mu_I(N) = 0$ such that $b(x) \in J$ whenever $x \notin N$.

Proof. For (a), let J support E_τ . Suppose $b(x) = i \notin J$ and $a(x) = c$. Choose $d \neq c$. Blindness gives the same output (i, c) on $x^{i \leftarrow c}$ and $x^{i \leftarrow d}$. Exactly one of these configurations succeeds. But they agree on J , contradicting support of the success event.

For (b), choose $E_0, N \in \Sigma_I$ with $\mu_I(N) = 0$ and $E_\tau \triangle E_0 \subseteq N$. Let J support both E_0 and N . If $x \notin N$ and $b(x) = i \notin J$, the two replacements from the preceding argument also lie outside N . They have equal membership in E_0 , hence in E_τ , again a contradiction. $\square$

Remark 5.2 (Two nonconverses). A map supported by countably many input coordinates can have uncountable output range. Conversely, countable range of targets alone does not establish measurability of its target/guess fibers. To deduce fair probabilities by summing (3), the fibers must be measurable as well. The theorem should not be paraphrased as “measurable success alone implies probability $1/q$.”

6 A continuous selector with maximally nonmeasurable success

In this section take $q = 2$. Let $C = 2^{\mathbb{N}}$ be Cantor space and take the tagged disjoint union

$$I = \mathbb{N} \sqcup C.$$

The distinction between the two tags is part of the definition. For a configuration x , read its natural-number coordinates and form $r = x \upharpoonright \mathbb{N} \in C$. Select the box on the *other* side indexed by r , and guess zero:

$$b(x) = r \in C, \quad a(x) = 0. \quad (5)$$

This is a legal strategy when countably many boxes may be opened in one move. It is blind. Its output map is cylinder-measurable; with the product topology on the domain and the usual disjoint-union topology on I , it is even continuous.

Its success set is

$$E = \{x : x_{x \upharpoonright \mathbb{N}} = 0\},$$

where the outer evaluation uses the C -tagged coordinate.

Theorem 6.1 (Inner measure zero, outer measure one). *For the diagonal strategy,*

$$\mu_*(E) = 0, \quad \mu^*(E) = 1.$$

In particular, E belongs neither to Σ_I nor to its completion.

Proof. Let $A \in \Sigma_I$ with $A \subseteq E$, and let K be a countable support for A . For any $x \in A$, its selected label $r = x \upharpoonright \mathbb{N}$ must belong to $K \cap C$. Otherwise changing only the r -tagged coordinate from zero to one leaves A unchanged but destroys success. Thus

$$A \subseteq \{x : x \upharpoonright \mathbb{N} \in K \cap C\}.$$

A countable set of fair infinite binary sequences has probability zero, so $\mu_I(A) = 0$. This proves the inner-measure assertion. Repeating the argument for a measurable subset of E^c , now changing the selected bit from one to zero, shows $\mu_*(E^c) = 0$. The outer-measure assertion follows by complementation. $\square$

6.1 Every probability extension after adjoining success

Write $\Sigma_I[E] = \sigma(\Sigma_I, E)$. Its members have the form

$$(A \cap E) \cup (B \cap E^c), \quad A, B \in \Sigma_I. \quad (6)$$

This is a σ -algebra directly: take complements and countable unions in the two components.

Theorem 6.2 (Classification by a measurable density). *For every measurable $h : X_I \rightarrow [0, 1]$, the formula*

$$\nu_h((A \cap E) \cup (B \cap E^c)) = \int_A h d\mu_I + \int_B (1 - h) d\mu_I \quad (7)$$

defines a countably additive probability extending μ_I . Every countably additive extension to $\Sigma_I[E]$ is obtained this way, with h unique up to μ_I -almost-everywhere equality.

Proof. Suppose two pairs (A, B) and (A', B') represent the same set. Then $A \triangle A'$ is a measurable subset of E^c , and $B \triangle B'$ is a measurable subset of E . Both are null by Theorem 6.1, so the formula is well-defined.

For a disjoint sequence of represented sets, the corresponding A components are pairwise disjoint modulo null sets, as are the B components. Integration of h and $1 - h$, or monotone convergence after removal of those countably many null overlaps, proves countable additivity. Taking $A = B = X_I$ gives total mass one. Taking $A = B = D$ shows $\nu_h(D) = \mu_I(D)$ for $D \in \Sigma_I$.

Conversely, let ν be an extension and define $\lambda(A) = \nu(A \cap E)$ on Σ_I . This is a measure and $0 \leq \lambda \leq \mu_I$. The Radon–Nikodym theorem yields a measurable density $0 \leq h \leq 1$ with $\lambda(A) = \int_A h d\mu_I$. Since $\nu(B \cap E^c) = \mu_I(B) - \lambda(B)$, formula (7) follows. Uniqueness is the usual almost-everywhere uniqueness of that density. $\square$

For constant $h = \theta$, the extension assigns $\nu_h(E) = \theta$ for any $\theta \in [0, 1]$. Every fixed finite collection of boxes still consists of independent fair bits. Thus those fixed-coordinate laws alone do not determine the probability of a dynamically indexed success event. The fair extension of Theorem 4.1 supplies one consistent answer; it does not turn that answer into a consequence of the original cylinder law alone.

7 Finite output maps and executable decision trees

Now let $I = \{0, \dots, n-1\}$ and first take $Q = \{0, 1\}$. The cube graph Q_n has vertex set Q^n and edges joining configurations differing in one coordinate. A perfect matching partitions its vertices into edges.

Proposition 7.1 (Blind maps are labeled matchings). *Blind binary output maps on n boxes are in bijection with perfect matchings of Q_n whose edges each carry an arbitrary label in $\{0, 1\}$. If M_n is the number of perfect matchings, the number of blind output maps is*

$$B_n = 2^{2^{n-1}} M_n.$$

Every such map succeeds at exactly 2^{n-1} configurations.

Proof. If $\tau(x) = (i, a)$, blindness pairs x with the configuration obtained by flipping its i th coordinate, giving both vertices the same target direction i and guess label a . This pairing is an involution without fixed points, hence a perfect matching. Conversely, a matching and its labels specify a blind output map by using the edge direction as target and its label as guess. Each edge has exactly one endpoint whose target bit equals its label. There are 2^{n-1} edges. $\square$

This is a statement about extensional output maps, not about descriptions of programs. Different decision trees can compute the same map.

Definition 7.2 (Recursively sliceable matching). The unique matching of a one-dimensional cube is *sliceable*. For $n \geq 2$, a matching is sliceable if there is a coordinate k such that it has no edge in direction k and the induced matchings in both faces $x_k = 0$ and $x_k = 1$ are sliceable. Coordinates retain their labels during restriction.

Theorem 7.3 (Exact executability criterion). *A blind binary output map on finitely many boxes is realizable by a deterministic decision tree that never queries its eventual target if and only if its underlying matching is sliceable. The choice of guess labels does not affect realizability.*

Proof. A sliceable matching supplies a decision tree recursively: query a slicing coordinate and continue in the observed face. At a final one-dimensional face, its unique coordinate is still unread; output that coordinate and the label of its edge. This works for every assignment of labels.

Conversely, start with a legal decision tree. Repeated queries can be deleted. If a leaf is reached with more than one unread coordinate, its target is one of them. Refine that leaf by querying all the other unread coordinates and leave the target unread. Its prescribed output remains unchanged. The resulting tree ends at one-dimensional faces. Its first query k can be no eventual target, so the matching has no edge in direction k . The two restricted trees realize the induced face matchings. Induction on dimension proves sliceability.

A tree that initially makes no query is handled by the same refinement: its constant target is left unread while the other coordinates are queried. Thus the argument covers early stopping as well as full-depth trees. $\square$

Lemma 7.4 (Restriction lemma). *If a sliceable matching has no edges in any of a specified set K of coordinate directions, its restriction to every face obtained by fixing the coordinates of K is sliceable, provided that face has positive dimension.*

Proof. Use its legal decision tree and substitute the fixed values for queries in K , deleting those query nodes. Since none of these coordinates is ever a target, the resulting restricted tree is still legal. Apply Theorem 7.3. $\square$

7.1 The first failure of the blindness converse

For $n \leq 3$, every cube matching is sliceable. For $n = 2, 3$, the number of matching edges crossing each coordinate cut is even: each half contains 2^{n-1} vertices, and internal matching edges cover them in pairs. A matching of Q_2 has two edges and one of Q_3 has four. Consequently they cannot use every coordinate direction. Split along a missing direction and proceed inductively.

At dimension four consider the matching below. Vertices are shown as integers; coordinate i is bit i , with bit zero least significant.

Edge	(0, 1)	(2, 3)	(4, 6)	(5, 13)
Direction	0	0	1	3
Edge	(7, 15)	(8, 12)	(9, 11)	(10, 14)
Direction	3	2	1	2

Every vertex occurs exactly once, and all four coordinate directions occur. Label every edge zero. The resulting map is blind but cannot be executed: whatever the first queried coordinate is, it is the final target on

some configuration. Making no query cannot implement a map whose target varies. This proves a sharp four-box separation, independently of the enumeration below.

8 Exact enumeration of executable maps

Let F_n count sliceable matchings of the labeled cube Q_n . Put $F_0 = 0$ as an auxiliary convention and $F_1 = 1$.

Theorem 8.1 (Inclusion–exclusion recurrence). *For every $n \geq 2$,*

$$F_n = \sum_{k=1}^{n-1} (-1)^{k+1} \binom{n}{k} F_{n-k}^{2^k}. \quad (8)$$

The number of distinct legal binary output maps is

$$L_n = 2^{2^{n-1}} F_n. \quad (9)$$

Proof. For each coordinate i , let A_i be the class of matchings that can be sliced first in direction i . Then $F_n = |\bigcup_i A_i|$. For a set K of $k < n$ possible first coordinates, the intersection $\bigcap_{i \in K} A_i$ consists exactly of independent choices of sliceable matchings on the 2^k faces obtained by fixing K .

One direction follows from Lemma 7.4. In the other direction, given the matchings on those faces, one may first query any chosen coordinate of K , query the remaining coordinates of K , and then execute the supplied face tree. Thus

$$\left| \bigcap_{i \in K} A_i \right| = F_{n-k}^{2^k}.$$

The intersection for all n coordinates is empty because no matching edge direction could remain. Inclusion–exclusion proves (8). Arbitrary edge labels are permitted by Theorem 7.3, proving (9). $\square$

n	F_n	M_n	Legal output maps L_n	Blind output maps B_n
1	1	1	2	2
2	2	2	8	8
3	9	9	144	144
4	232	272	59 392	69 632
5	206 065	589 185	$2^{16} F_5$	$2^{16} M_5$
6	212 181 312 096	—	$2^{32} F_6$	—

The M_n entries agree with A005271 [5] and were recomputed here by exact enumeration or a permanent dynamic program through $n = 5$. The row $n = 6$ intentionally does not import an unverified larger enumeration into the certificate calculation.

[write] Added 3 October 2026, batch 87. A005271 records $M_6 = 16\,332\,454\,526\,976$ (read at intake). It is not used anywhere in this report; it is consistent with $D_6 \leq M_6$ for the count D_6 of (25) (their ratio is about 0.1226). The column F_n and the ternary values below are not in the OEIS (Section 1.4).

For $n = 4$, a matching is nonsliceable precisely when it uses all four directions, since every three-dimensional matching is sliceable. Inclusion–exclusion gives the number of such matchings as

$$272 - 4 \cdot 9^2 + 6 \cdot 2^4 - 4 \cdot 1^8 = 40.$$

Hence exactly $40 \cdot 2^8 = 10\,240$ blind four-box output maps are not executable.

8.1 The q -ary version

For a general alphabet Q , replace cube edges by full coordinate lines of length q . Blind output maps correspond to partitions of Q^n into such lines with one guess label per line. The decision-tree proof is unchanged, with q children at each query. If $F_n^{(q)}$ counts recursively sliceable line partitions, then

$$F_1^{(q)} = 1, \quad F_n^{(q)} = \sum_{k=1}^{n-1} (-1)^{k+1} \binom{n}{k} \left(F_{n-k}^{(q)}\right)^{q^k}. \quad (10)$$

There are $q^{q^{n-1}} F_n^{(q)}$ legal output maps. In particular, the ternary values begin

$$1, \quad 2, \quad 21, \quad 33976, \quad 188162675402345.$$

This counts maps, not trees, so alternative query orders are not counted multiple times.

9 All-orders asymptotics for sliceable matchings

We now concentrate on the binary sequence F_n . Its doubly exponential scale is natural because an initial query creates two independent subproblems. The inclusion–exclusion terms encode overlaps between alternative initial queries.

Theorem 9.1 (Full inverse-power expansion). *There is a positive constant*

$$\gamma = \lim_{n \rightarrow \infty} 2^{-n} \log F_n = \sum_{k=2}^{\infty} 2^{-k} \log r_k, \quad r_k = \frac{F_k}{F_{k-1}^2}, \quad (11)$$

such that, for every integer $N \geq 1$, there are rational coefficients $c_1, \dots, c_N$ satisfying

$$\log F_n = \gamma 2^n - \log n + \sum_{j=1}^N \frac{c_j}{n^j} + O(n^{-N-1}). \quad (12)$$

The coefficients are recursively computable. The first terms are

$$\log F_n = \gamma 2^n - \log n - \frac{3}{2n} + \frac{25}{8n^2} - \frac{27}{4n^3} + O(n^{-4}), \quad (13)$$

$$F_n = \frac{e^{\gamma 2^n}}{n} \left(1 - \frac{3}{2n} + \frac{17}{4n^2} - \frac{12}{n^3} + O(n^{-4}) \right). \quad (14)$$

9.1 Ratio bounds and existence of the growth constant

By the union bound and the first two Bonferroni bounds for the sets A_i in the proof of Theorem 8.1,

$$nF_{n-1}^2 - \binom{n}{2} F_{n-2}^4 \leq F_n \leq nF_{n-1}^2. \quad (15)$$

Also $F_n \geq F_{n-1}^2$ by choosing one specified first coordinate. Thus $1 \leq r_n \leq n$. In fact,

$$\frac{n}{2} \leq r_n \leq n \quad (n \geq 2), \quad \frac{r_n}{n} \geq 1 - \frac{2}{n-1} \quad (n \geq 5). \quad (16)$$

The first inequality is checked directly at $n = 2, 3, 4$. If it holds at $n - 1$, then (15) gives

$$r_n \geq n \left(1 - \frac{n-1}{2r_{n-1}^2} \right) \geq n \left(1 - \frac{2}{n-1} \right) \geq \frac{n}{2}$$

for $n \geq 5$, proving the induction and the second assertion.

Since

$$2^{-n} \log F_n - 2^{-(n-1)} \log F_{n-1} = 2^{-n} \log r_n,$$

the scaled logarithms increase and are bounded by the convergent series $\sum_{n \geq 2} 2^{-n} \log n$. This proves (11); positivity follows already from $F_2 = 2$. It also gives the exact tail identity

$$\gamma 2^n - \log F_n = \sum_{j=1}^{\infty} 2^{-j} \log r_{n+j}. \quad (17)$$

9.2 The first correction terms

The next Bonferroni term bounds the error in (15) by $\binom{n}{3} F_{n-3}^8$. Dividing by $n F_{n-1}^2$ and using (16) yields

$$\frac{r_n}{n} = 1 - \frac{n-1}{2r_{n-1}^2} + O(n^{-4}). \quad (18)$$

For clarity, the normalized triple-intersection term is

$$\frac{\binom{n}{3}}{n r_{n-1}^2 r_{n-2}^4} = O(n^{-4}).$$

Starting with $r_n = n + O(1)$ from (16), repeated substitution into (18) improves the approximation one power at a time. It gives

$$r_n = n - \frac{1}{2} - \frac{1}{n} - \frac{23}{8n^2} + O(n^{-3}), \quad (19)$$

$$\log(r_n/n) = -\frac{1}{2n} - \frac{9}{8n^2} - \frac{41}{12n^3} + O(n^{-4}). \quad (20)$$

For example, writing $r_{n-1} = n - \frac{3}{2} - n^{-1} + O(n^{-2})$ gives

$$\frac{n-1}{2r_{n-1}^2} = \frac{1}{2n} + \frac{1}{n^2} + \frac{23}{8n^3} + O(n^{-4}),$$

which displays the coefficient that is most easily missed.

Substitute (20) into (17). The geometric moments

$$\sum_{j \geq 1} \frac{1}{2^j} = 1, \quad \sum_{j \geq 1} \frac{j}{2^j} = 2, \quad \sum_{j \geq 1} \frac{j^2}{2^j} = 6, \quad \sum_{j \geq 1} \frac{j^3}{2^j} = 26$$

give

$$\gamma 2^n - \log F_n = \log n + \frac{3}{2n} - \frac{25}{8n^2} + \frac{27}{4n^3} + O(n^{-4}).$$

This proves (13). Exponentiating gives (14).

9.3 A coefficient algorithm and proof to every order

Put $z = 1/n$. There is a unique formal power series $R(z) = 1 + \sum_{j \geq 1} a_j z^j$ satisfying

$$R(z) = \sum_{k \geq 1} \frac{(-1)^{k+1}}{k!} z^{2^k - k - 1} \frac{\prod_{h=0}^{k-1} (1 - hz)}{\prod_{j=1}^{k-1} \left[(1 - jz) R\left(\frac{z}{1 - jz}\right) \right]^{2^j}}. \quad (21)$$

The $k = 1$ term is 1. Every other term is divisible by z , and its coefficient of degree d involves coefficients of R of degree at most $d - 1$. Also $2^k - k - 1 \rightarrow \infty$, so every coefficient receives contributions from only finitely many k . These observations prove existence, uniqueness, and rational computability coefficient by coefficient.

To see that this is an asymptotic rather than merely a formal calculation, divide the exact recurrence by nF_{n-1}^2 . For a fixed k , its normalized term is

$$\frac{\binom{n}{k}}{n \prod_{j=1}^{k-1} r_{n-j}^{2^j}} = O\left(n^{k+1-2^k}\right). \quad (22)$$

The Bonferroni inequalities sandwich the exact answer between any two consecutive inclusion–exclusion truncations. Their difference is precisely the next intersection term, bounded by (22). Given any desired power of $1/n$, choose a fixed truncation long enough that this error is smaller. In the remaining finite expression, the dependence on an error in each shifted ratio is multiplied by at least one extra power of $1/n$. Starting with $r_n/n = 1 + O(n^{-1})$, induction therefore improves the error by one power at each step. This proves

$$r_n/n = R(1/n) + O(n^{-N-1})$$

when R is truncated at degree N .

Finally define the formal series

$$C(z) = - \sum_{j \geq 1} 2^{-j} \left[\log(1 + jz) + \log R\left(\frac{z}{1 + jz}\right) \right]. \quad (23)$$

Each coefficient is a rational linear combination of geometric moments and is therefore well-defined and rational. Its coefficients are the c_j in (12). To justify termwise asymptotic expansion in (17), split the sum at $j = n/2$. In the initial part, Taylor remainders are bounded by a constant times $(1 + j)^{N+1} n^{-N-1}$ and are summable against 2^{-j} . The remaining tail is exponentially small, using $r_{n+j} \leq n + j$. This completes the all-orders proof.

The exponents at which successive intersection sizes first enter are

$$2^k - k - 1 = 0, 1, 4, 11, 26, \dots$$

Thus a high asymptotic order needs surprisingly few intersection types. This is an all-orders Poincaré expansion on a doubly exponential scale; no claim about beyond-all-orders terms is being made.

9.4 Effective bounds for the constant

For $N \geq 4$, let $g_N = 2^{-N} \log F_N$. From (16), (17), and concavity of the logarithm,

$$\begin{aligned} g_N + 2^{-N} \left[\log(N+1) + \log\left(1 - \frac{2}{N}\right) \right] &\leq \gamma \\ &\leq g_N + 2^{-N} \left[\log(N+1) + \frac{1}{N+1} \right]. \end{aligned} \quad (24)$$

For the upper bound, use $\log(N + j) \leq \log(N + 1) + (j - 1)/(N + 1)$ and sum. For the lower bound, use $r_{N+j} \geq (N + j)(1 - 2/(N + j - 1)) \geq (N + 1)(1 - 2/N)$.

The exact recurrence through $N = 20$ gives rounded evaluations

$$0.4381576906086384 \lesssim \gamma \lesssim 0.4381578365013189.$$

The proved statement is the symbolic enclosure (24); these displayed decimals are rounded evaluations, not an interval-arithmetic certificate. A rough corresponding base is $e^\gamma \approx 1.54985$. The supplied checker records the exact recurrence data and the formulas used for the enclosure.

[write] Added 3 October 2026, batch 87. The checker's recorded run, `data/verification_report.json`, gives the two sides of (24) at $N = 20$ as 65-digit rounded evaluations,

$$0.43815769060863840388 \dots \quad \text{and} \quad 0.43815783650131883608 \dots$$

The two 16-digit decimals displayed above are outward roundings of these (the lower one rounded down, the upper one rounded up in its last digit). They remain rounded evaluations, not an interval-arithmetic certificate.

10 Executable blind maps are doubly exponentially rare

A small finite matching count already combines with the preceding analysis to give a uniform rarity theorem. Let D_6 count matchings of Q_6 that omit at least one coordinate direction. A prescribed set of k omitted directions separates the graph into 2^k independent copies of Q_{6-k} , so

$$\begin{aligned} D_6 &= \sum_{k=1}^5 (-1)^{k+1} \binom{6}{k} M_{6-k}^{2^k} \\ &= 2\,001\,589\,252\,896. \end{aligned} \tag{25}$$

This calculation uses only $M_1, \dots, M_5$, all recomputed by the accompanying exact program.

Theorem 10.1 (Explicit rarity bound). *For $n \geq 6$, the fraction of blind binary output maps that are executable satisfies*

$$\frac{L_n}{B_n} = \frac{F_n}{M_n} < \frac{2}{n+1} \left(\frac{10\,396\,884\,292\,704}{12\,009\,535\,517\,376} \right)^{2^{n-6}}. \tag{26}$$

In particular, this fraction tends to zero doubly exponentially.

Proof. Partition Q_n into 2^{n-6} six-dimensional faces and choose independently a matching counted by D_6 in each face. This gives $M_n \geq D_6^{2^{n-6}}$.

From $r_{n+j} \geq (n + j)/2$ and (17),

$$F_n \leq \frac{2}{n+1} e^{\gamma 2^n}.$$

Use the upper bound in (24) at $N = 6$ and the elementary strict inequality $e^{1/7} < 7/6$ to obtain

$$e^{64\gamma} \leq 7F_6 e^{1/7} < \frac{49F_6}{6}.$$

Thus

$$\frac{F_n}{M_n} < \frac{2}{n+1} \left(\frac{49F_6}{6D_6} \right)^{2^{n-6}}.$$

The numerator and denominator in (26) are exactly $49F_6$ and $6D_6$. Their strict order is an integer comparison, verified by the program. No numerical approximation to γ is needed. $\square$

This theorem concerns the uniform counting measure on extensional blind maps. It is not a claim about the frequency with which a particular programming language generates legal programs. It quantifies how weak the semantic blindness condition becomes as a substitute for actual executability.

11 How much does a coordinate budget reveal?

Return to arbitrary finite q , but now assume that I is countable. This removes the variable-evaluation measurability obstruction. Let $\tau = (b, a)$ be blind and measurable, and regard Q as the cyclic group $\mathbb{Z}/q\mathbb{Z}$. Define the residual

$$Z = x_{b(x)} - a(x) \pmod{q}.$$

For $K \subseteq I$, let X_K denote all values on K and put $\rho_K = \mu_I(b \in K)$. Entropies use natural logarithms.

Theorem 11.1 (Sharp information-budget inequalities). *The random variable Z is uniform on Q . For every measurable estimator $g(X_K)$,*

$$\mathbb{P}(g(X_K) = Z) \leq \frac{1}{q} + \left(1 - \frac{1}{q}\right) \rho_K, \quad (27)$$

$$H(Z \mid X_K) \geq (1 - \rho_K) \log q, \quad (28)$$

$$I(Z; X_K) \leq \rho_K \log q. \quad (29)$$

For $q = 2$ and every event A determined by X_K ,

$$\mu_I(E_\tau \triangle A) \geq \frac{1}{2} \mu_I(b \notin K). \quad (30)$$

Proof. For a fixed $i \notin K$ and $c \in Q$, the fiber $\{b = i, a = c\}$ is measurable and independent of the value at i in the pointwise invariance sense. Intersecting it with an event determined by X_K preserves that invariance. Integration in coordinate i therefore gives, for every $z \in Q$ and every such event D ,

$$\mathbb{P}(D, b = i, a = c, Z = z) = \frac{1}{q} \mathbb{P}(D, b = i, a = c).$$

Sum over $i \notin K$ and c . Conditional on X_K , the contribution of the external-target case is exactly a uniform subprobability of mass $1 - p(X_K)$, where

$$p(X_K) = \mathbb{P}(b \in K \mid X_K).$$

Consequently the conditional law of Z has the form

$$(1 - p(X_K)) U_q + p(X_K) \eta_{X_K},$$

where η_{X_K} is some probability on Q whenever $p > 0$. Each point mass is at most $(1 - p)/q + p$, proving (27). Concavity and nonnegativity of finite entropy give conditional entropy at least $(1 - p) \log q$, proving (28) after integration.

Taking $K = \emptyset$ proves that Z is uniform, so $H(Z) = \log q$ and (29) follows. When $q = 2$, on the external-target case the conditional probability of either success or failure is $1/2$. Hence

$$\mu_I((E_\tau \triangle A) \cap \{b \notin K\}) = \frac{1}{2} \mu_I(b \notin K),$$

which implies (30). □

Example 11.2 (Sharpness). Take two data coordinates $i_{\text{in}} \in K$ and $i_{\text{out}} \notin K$, plus a control block inside K disjoint from both. The player reads only the control block. On a control event of probability ρ , it chooses i_{in} ; otherwise it chooses i_{out} . It always guesses zero. Conditional on X_K , the residual is known exactly on the first case and is uniform on the second. Thus all three bounds (27)–(29) are attained. A countable fair control block permits any $\rho \in [0, 1]$, by comparison of its base- q real with a threshold; finite control blocks give the corresponding q -adic probabilities.

These inequalities quantify an observer’s information about the outcome, not the player’s information while running the strategy. In the example, the observer’s set K contains a coordinate that the player deliberately leaves unopened. For finite K , (30) gives a direct lower bound on the error of any finite-coordinate approximation to the success event.

12 Effective finite certificates and the Busy Beaver barrier

12.1 Finite averaging without measure theory

For a blind map on Q^n , its fibers are unions of full lines in the selected coordinate, with one successful point per q -point line. Therefore

$$\sum_{x \in Q^n} \mathbf{1}_{E_\tau}(x) = q^{n-1}. \quad (31)$$

For m players, the total of all scores is mq^{n-1} . Some configuration has score at most $\lfloor m/q \rfloor$. This is an integer identity amenable to a small exact checker.

A finite output-table certificate consists of q, n, m , the output pair for each player on each configuration in lexicographic order, and optionally a proposed counterexample configuration. The checker verifies dimensions and ranges, condition (1) for every one-coordinate replacement, and the exact score. A theorem connecting accepted tables to actual programs is a separate obligation: blind tables alone need not be executable, as the four-box example demonstrates.

12.2 Total oracle strategies have finite certificates

Theorem 12.1 (Uniform extraction under a totality promise). *Given programs for finitely many deterministic oracle strategies on $Q^\mathbb{N}$, promised to halt on every oracle and never to query their eventual targets, there is a uniform partial algorithm that halts on every such promised input and returns a finite prefix σ whose every extension gives team score at most $\lfloor m/q \rfloor$.*

Proof. First prove existence of a finite common horizon. Every individual halting run reads finitely many coordinates, has a finite running time, and outputs a finite target index. All oracles agreeing on its queried coordinates produce the same run and output. These finite-coordinate neighborhoods cover $Q^\mathbb{N}$. Compactness supplies a finite subcover, and hence finite common bounds on queried indices, target indices, and running time. Repeat for the finitely many players and take maxima.

The bounds can be found effectively under the promise. Search over pairs (L, T) of positive integers. For each of the q^L prefixes, simulate every player for at most T steps. Reject the candidate bounds if any simulation asks for a coordinate outside $\{0, \dots, L-1\}$, fails to halt within T , or outputs a target at least L . A sufficiently large pair passes by the preceding argument.

The accepted simulations produce finite output tables on Q^L . They are blind because the original programs are legal. By (31), exhaustive search finds a prefix with score at most $\lfloor m/q \rfloor$. Every extension produces the same runs and the same target values, since all targets lie within the prefix. Return that prefix. $\square$

The promise is substantive. This theorem is not a decision procedure for whether arbitrary oracle programs are total or legal. It also makes no useful polynomial or elementary running-time claim.

Theorem 12.2 (No computable length bound for the horizon). *For any effective programming formalism supporting ordinary Turing simulation, no total computable function of program length bounds the necessary input horizon of every total legal oracle strategy. This remains true for strategies that make no oracle queries.*

Proof. Given a program P , compile a strategy S_P that simulates P on blank input, counts its steps, and, if P halts after t steps, outputs target $t + 1$ and guess zero. The strategy makes no queries. Whenever P halts, S_P is total and legal, and a horizon including its target must be at least $t + 2$.

Suppose a total computable bound f existed. The length of the compiled program S_P is computable from P . If P halts, it must do so before the computable bound $f(|S_P|)$. Simulate P for that many steps: a failure to halt would establish that it never halts. This would decide the halting problem, a contradiction. $\square$

For an encoding equipped with an additive-overhead simulation compiler, this argument can be expressed as

$$H(n + c) \geq \text{BB}_{\text{time}}(n) + 2,$$

where H is the maximum horizon over total strategies of length at most its argument and BB_{time} is a *program-length running-time* Busy Beaver function. The compiler and encoding are hypotheses of this sharpened comparison. This is not the same numerical function as an n -state Rado printing-score Busy Beaver function. The distinction mirrors the explicit separation between machine semantics, maxima, and compiler assumptions in the inspected ProveIt core [7].

[write] **Added 3 October 2026, batch 87.** The Busy Beaver development in Computability/BusyBeaver formalizes blank-tape two-symbol machines, their halting scores and the domination theorem, with the compiler as a separately stated hypothesis; it says nothing about H or BB_{time} , and neither Theorem 12.1 nor Theorem 12.2 is formalized there or elsewhere in ProveIt. The Hilbert’s-tenth review cited in Section 1.4 read this section and agrees with its hypotheses: the promise in Theorem 12.1 is essential, and the displayed inequality is conditional on the encoding and the compiler.

13 Executed checks and reproducibility

The accompanying program `verify_certificates.py` uses only the Python standard library. It validates the four-box nonrealizable matching, recomputes every cube matching through dimension four, checks sliceability independently against the recurrence, and recomputes M_5 by an exact permanent dynamic program.

For the latter, bipartition the cube into even- and odd-parity vertices. Process even vertices in a fixed order, and use the bit mask of already matched odd vertices as a state. The recurrence sums over unused adjacent odd vertices. At dimension five there are only 2^{16} possible masks, rather than a search through all permutations. The result is $M_5 = 589185$.

The program also enumerates all $4^4 = 256$ two-box output tables, finding exactly eight blind ones. For all ordered teams of one through four such tables it checks (31) and the sharp minimax bound. The resulting counts by minimum score are:

Players	Ordered teams	Minimum 0	Minimum 1	Minimum 2
1	8	8	0	0
2	64	52	12	0
3	512	236	276	0
4	4096	988	2784	324

An additional explicit three-box team has player p target coordinate p and guess $1 - x_{p+1 \bmod 3}$. Each player reads one other coordinate. Its score is zero on 000 and 111, and two on the other six configurations. Thus its probability of at least two successes is $3/4$, attaining the first-moment upper bound $(3/2)/2$. This average-case fact does not contradict its failure to guarantee two successes.

The checker further validates 1,320 balanced-list constructions: $2 \leq q \leq 12$, $1 \leq r < q$, and $1 \leq m \leq 20$. It computes F_n through $n = 20$, verifies the exact integer comparison used in Theorem 10.1, and records rounded evaluations of (24). The separate script `formal_series.py` verifies the formal fixed-point identity and computes the coefficients of (21) and (23) through degree ten, using exact rational arithmetic. It independently checks the displayed coefficients; the next multiplicative coefficient after those in (14) is $2677/48$.

Verification boundary. All these finite tests were executed successfully. They are exact arithmetic checks except for explicitly labeled decimal evaluations of logarithms. They are not a proof-assistant verification of the infinite theorems, nor do they replace the proofs of the recurrence and asymptotic remainder estimates.

[write] **Added 3 October 2026, batch 87.** In the collection the programs are shipped as `code/verify_certificates.py` and `code/formal_series.py` (with the delivered `code/Makefile`), and their recorded outputs as `data/certificates.json`, `data/verification_report.json` and `data/formal_series_report.json`, all byte-identical to the delivery. Both programs write into the current directory, and the first one rewrites `certificates.json` too, so they are rerun on a scratch copy only; the report's `README.md` gives the commands. The intake's replay on such a copy (3 October 2026, Python 3.14.4 on Windows) passed all three commands of the `Makefile` target `verify` in a few seconds, and the three regenerated files equal the shipped ones after the CRLF line ends that Windows text mode writes are stripped.

14 Foundations and a concrete ProveIt integration plan

14.1 Set-sized objects and explicit ambient axioms

All configuration spaces in this note are sets Q^I for set-sized I . The index space $\mathbb{N} \sqcup 2^{\mathbb{N}}$, the set of strategies, and the σ -algebras used in the proofs are sets. No class of all ordinals, all surreal numbers, or all sets is being treated as a measurable sample space.

The measure-theoretic development is given in ordinary classical mathematics, with ZFC as a sufficient ambient theory. The uses of countable support, product probability, null-set completion, Radon–Nikodym, and compactness are stated explicitly. No equivalence with ZF, ZF plus dependent choice, or a subsystem of arithmetic has been proved here. In particular, the proof should not be transported into a choiceless model merely by reusing the same notation.

The finite counting and certificate results do not use an uncountable probability space. Their proofs employ finite sets, induction, and integer arithmetic. The asymptotic theorem adds real analysis of an explicit integer recurrence. These layers can therefore be formalized separately, instead of requiring the entire continuum theory before checking a finite certificate.

14.2 Relevant inspected repository interfaces

The pinned Busy Beaver core explicitly separates a Rado-style machine representation and score maxima from the compiler interface used in domination arguments [7]. This is the right model for Theorem 12.2: first state which programming model and size measure are used, then prove or assume the appropriate compilation theorem.

The finite-list-coding README documents validity guards, unique decoding, round-trip properties, and genuine first-order arithmetic predicates [8]. These are suitable interfaces for encoding output tables and finite computation histories. A malformed table must not become a meaningful object just because a total decoder supplies a default value.

The closure-axiomatization README distinguishes semantic directions from a deductive-equivalence layer built on a first-order calculus [9]. That distinction is relevant to the eventual weak-foundation audit: a theorem proved in an ambient Lean type theory is not automatically a theorem that a specified first-order set theory proves internally. These are descriptions of inspected source interfaces, not a claim that the entire repository was rebuilt during this task.

14.3 Suggested modules and proof obligations

A practical implementation would start with four small mathematical modules rather than a monolithic formalization.

Finite blind maps. Define a table on functions $\text{Fin}(n) \rightarrow \text{Fin}(q)$, a coordinate-update operation, and predicate (1). Prove the finite identity (31), then its team bound. A Boolean checker needs a reflection theorem equating acceptance with the mathematical predicate.

Decision trees and partitions. Give queries and leaves an operational semantics with an explicit set of previously queried coordinates. Prove that legal executions induce blind maps. Formalize the matching or line-partition correspondence, the leaf-refinement argument, and the sliceability equivalence. The four-box example is a regression test against accidentally declaring the converse of blindness to be automatic.

Enumeration and asymptotics. Prove the intersection formula for first-query classes before deriving inclusion–exclusion. Treat F_n as the cardinality of an explicit finite family, then prove the recurrence agrees with the numerical evaluator. The exact rarity bound can be checked before the all-orders asymptotic theorem, since it needs only ratio inequalities and finite constants.

Countable support and probability. State the measurable codomain assumptions explicitly, establish the countable-support factorization, and prove finite-coordinate integration lemmas. The fresh replacement construction then supplies a measurable-space extension with a pushforward measure. The diagonal counterexample should be formalized before any automatic rule for evaluating a process at a random index is trusted.

No Lean source claiming these new theorems as already established is included. The supplied code is a working computational companion, while this section specifies the remaining proof-assistant work.

15 Further research questions and topics

The first question below is a question from Glazer’s paper. The others are precisely formulated directions suggested by this note; they are not all claimed to be previously recognized open problems.

- Q1. The unrestricted choiceless continuum problem.** Determine the ZF status of the three-player binary continuum game posed in [1]. A relevant next step is to compare any proposed strategy with the exact output-regularity obstruction here, without imposing measurability by assumption.

- Q2. Larger measurable structures.** Which versions of Theorem 3.1 remain true when cylinder measurability is replaced by full product-topology Borel measurability, the Baire property, or universal measurability? Each proposed replacement must specify the ambient measurable or topological structure on both configurations and target labels.
- Q3. Weak foundational strength.** Determine the weakest explicit choice and comprehension principles sufficient for the countable-support minimax proof. Separate a theorem about coded measurable maps from a theorem quantifying over all members of an abstract generated σ -algebra.
- Q4. Universal fair extensions.** Theorem 4.1 handles countably many output maps at once. Is there a useful maximal class of uncountably many maps whose success events admit a simultaneous countably additive fair extension? Which extra conditional-independence requirements obstruct such an extension?
- Q5. Joint extension classification.** For a finite team, characterize all extensions of the product measure to its success events, including their possible joint distributions. The single-event density classification in Theorem 6.2 suggests a finite-family theory with pointwise compatibility constraints.
- Q6. q -ary decision-tree asymptotics.** Starting from (10), derive a uniform-in- q expansion, identify the appropriate polynomial prefactor, and compare different alphabets. The fixed- q ratio method is a starting point, but uniformity in a growing alphabet is a separate problem.
- Q7. Beyond inverse powers.** Investigate the remainder after optimal truncation of (12), and whether the sparse intersection orders $2^k - k - 1$ govern identifiable smaller asymptotic scales. The all-orders result here does not settle such questions.
- Q8. Structural enumeration and database identification.** Identify the sliceable-matching sequence in the wider literature on decision-tree partitions, and determine whether an existing OEIS entry records it under another interpretation. Investigate symmetries, unlabeled versions, and counts refined by the set of permissible first queries.
- Q9. Optimal average-case teamwork.** For fixed (m, q, n) , determine the largest probability of at least t successes over legal decision trees. Compare it with the larger class of blind maps and locate the first parameter values where the optima differ. The exact 3/4 binary example gives a small benchmark.
- Q10. Complexity of realizability.** Given a succinct blind output map, what is the complexity of deciding whether it is executable? How does the answer change for an explicit truth table, a Boolean circuit, a matching list, or an oracle program with a supplied totality certificate?
- Q11. Sharper information costs.** Refine Theorem 11.1 when observation sets are adaptive, observation costs are nonuniform, or the box law has correlations. The product-law proof identifies precisely which missing-coordinate integration identity needs replacement.
- Q12. Kernel-checked certificate compilation.** Implement the finite reflection theorem and the decision-tree-to-table compiler, then connect the oracle horizon theorem to a formally specified machine model. Track compiler size and time overhead rather than asserting an encoding-independent Busy Beaver comparison.

[write] **Status of the questions, 3 October 2026, batch 87.** All twelve questions stay open as printed. **Q1** is Glazer’s question “Does ZF prove that $G([3], \mathbb{R}, \{0, 1\})$ is winnable?” from Section 5 of [1] (one arXiv version, v1 of 15 November 2022); the intake did not search the later literature for its status, and, as Appendix B says, this report does not

assert that it is unresolved everywhere. It is unrelated to the “Glazer’s Question 1” answered (as claimed, unrefereed) in Part VIII of the surreal report `polish-models-of-omnific-arithmetic`, which belongs to a different paper. For **Q8**, the database half was checked at intake: on 3 October 2026 no OEIS entry matched 1, 2, 9, 232, 206065 or 2, 21, 33976 (Section 1.4); identification in the wider literature on decision-tree partitions, and any OEIS submission, remain open.

16 Conclusion

Two distinct mistakes are exposed by this analysis. On uncountable products, a measurable choice of target does not ensure a measurable success event. On finite products, invariance under changing the target does not ensure that the output rule can be executed without querying it.

The first distinction leads to an exact measurable-output minimax theorem, a simultaneous fair extension, and a sharp countable-target obstruction. The second leads to a combinatorial classification, a closed recurrence for extensional legal rules, an all-orders asymptotic expansion, and a quantitative rarity theorem. These are complementary rather than competing explanations: regularity limits which infinite outcomes can be assigned probabilities, while executability limits which finite semantic maps are operationally meaningful.

The next verification milestone is not another numerical experiment. It is a kernel-checked bridge from a legal decision tree to an accepted finite table, followed by the sliceability and inclusion–exclusion theorems. That would turn the strongest finite results of this note into reusable components in `ProveIt`, while leaving the genuinely set-theoretic questions visible and separate.

A Assumption and claim ledger

Result	Main hypotheses	Verification and scope
Sharp minimax	Standard Borel targets; cylinder-measurable blind output maps; finite alphabet and team	Full proof; success-event measurability not assumed.
Fair extension	Same output regularity; countable team	Full construction; marginal fairness, not independence or uniqueness.
Target collapse	Blindness; cylinder-measurable success, or completed measurability	Full proof; output-map measurability not assumed.
Diagonal example	Tagged continuum of targets; cylinder product probability	Full proof of inner/outer measures and extension classification.
Sliceability	Finite deterministic query model; binary outputs	Full equivalence proof; finite counterexample checked.
Enumeration	Extensional output maps, not program descriptions	Recurrence proved; matching counts independently checked through dimension four.
All-orders expansion	Explicit recurrence; ordinary real analysis	Remainder argument supplied; rational coefficients computed separately.

Result	Main hypotheses	Verification and scope
Rarity	Exact counts $M_1, \dots, M_5$ and ratio inequalities	Counts recomputed; final comparison is exact integer arithmetic.
Oracle extraction	Totality and legality on every oracle are promises	Existence and uniform partial algorithm proved; no practical complexity bound.
Busy Beaver comparison	Specified encoding and simulation interface	Uncomputable horizon theorem proved; additive-overhead inequality conditional on its stated compiler.

B Source and priority audit

The mathematical anchor is the full text of [1], including its final continuum-box question, inspected also as a rendered PDF page. The original Łoś–Marczewski article was inspected as scanned pages; its general extension theory is credited, while the special density classification here is proved directly. The OEIS A005271 entry was inspected for the identity of the all-matching sequence, not as a substitute for the executed small counts.

Repository files were read at the pinned commit, especially the initial machine and compiler description in the Busy Beaver core, the finite-list coding README, and the closure-axiomatization README. No repository-wide build was run, and no repository files were modified.

The LinkedIn page was not fully accessible; research interests were instead anchored in authored papers and the author profile in [3]. A complete dissertation-level priority comparison was not obtained, and exact-sequence searches did not establish whether the sliceable sequence already has a separate catalog entry. Consequently this note makes no claim of exhaustive literature coverage, no assertion that its formulations are historically first, and no assertion that the original unrestricted problem remains unresolved everywhere as of the preparation date.

[write] Added 3 October 2026, batch 87. The intake read the arXiv record and PDF of [1] (one version; Section 1.4), the record of [2] (three versions, v1 of 19 December 2023 to v3 of 7 January 2024) and the entry A005271. The post cited as [3] was reachable at the cited samaritan-research.org address on 3 October 2026, with the title, authors, dates and Glazer’s author profile as cited; the same post is published on Epoch AI’s blog, <https://epoch.ai/blog/frontiermath-competition-setting-benchmarks-for-ai-evaluation>, which the batch-87 manuscript printed as Part VIII of polish-models-of-omnific-arithmetic cites; the intake did not establish which copy is the original. The remark above about a LinkedIn page names no address, and none is recorded here. The package’s checksum manifest SHA256SUMS (nine entries) was verified 9/9 at placement and is not shipped, by repository policy; the delivered PDF and README are not shipped either, and survive in the arrival commit’s archive. The priority search was not extended at intake.

References

- [1] Elliot Glazer, *A choiceless box game paradox*, arXiv:2211.10474, submitted 2022; inspected PDF dated January 9, 2023. <https://arxiv.org/abs/2211.10474>. [write, 3 October 2026: arXiv lists one version, v1 of 15 November 2022, 6 pages, math.LO and math.CO, DOI 10.48550/arXiv.2211.10474; the date January 9, 2023 is printed in arXiv’s PDF of v1.]

- [2] Elliot Glazer, *Global choice is not conservative over local choice for Zermelo set theory*, arXiv:2312.11902. <https://arxiv.org/abs/2312.11902>.
- [3] Tamay Besiroglu, Elliot Glazer, and Caroline Falkman Olsson, *FrontierMath competition: Setting benchmarks for AI evaluation*, January 16, 2025, updated March 18, 2025; author profile of Elliot Glazer. <https://samaritan-research.org/blog/frontiermath-competition-setting-benchmarks-for-ai-evaluation>.
- [4] J. Łoś and E. Marczewski, *Extensions of measure*, Fundamenta Mathematicae **36** (1949), 267–276. Catalog: <https://eudml.org/doc/213194>. Inspected scan: <https://scispace.com/pdf/extensions-of-measure-v0jwsbilmx.pdf>.
- [5] The OEIS Foundation, *A005271: Number of perfect matchings in n-cube*. <https://oeis.org/A005271>. Accessed October 3, 2026.
- [6] Vladimir Reshetnikov, *ProveIt*, repository snapshot 6fef5383b126be343cccc9baf47081ef37ab5afe. <https://github.com/VladimirReshetnikov/ProveIt>.
- [7] ProveIt contributors, *BusyBeaver/Core.lean*, at the snapshot in [6], particularly its introductory machine/maximum/compiler interface and coordinate read/write lemmas. <https://github.com/VladimirReshetnikov/ProveIt/blob/6fef5383b126be343cccc9baf47081ef37ab5afe/Computability/BusyBeaver/Lean/BusyBeaver/Core.lean>.
- [8] ProveIt contributors, *Coding finite lists in Peano arithmetic*, project README at the snapshot in [6]. <https://github.com/VladimirReshetnikov/ProveIt/blob/6fef5383b126be343cccc9baf47081ef37ab5afe/Logic/PeanoArithmetic/ListCoding/README.md>.
- [9] ProveIt contributors, *Closure axiomatization of set theory*, project README at the snapshot in [6]. <https://github.com/VladimirReshetnikov/ProveIt/blob/6fef5383b126be343cccc9baf47081ef37ab5afe/SetTheory/ClosureAxiomatization/README.md>.